

Olist E-Commerce: Marketplace Performance & Customer Experience Analysis

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Data Analytics Hackathon

Gradient Learnings

Marketplace Performance

Logistics Analysis

Customer Experience

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Executive Summary

Olist has experienced significant growth over a two-year period, effectively scaling its order volume and connecting small and medium Brazilian merchants to major e-commerce marketplaces. However, as the platform scaled, customer satisfaction, measured through 1-to-5-star review scores, became increasingly inconsistent.

This analysis leverages actual operational data spanning nearly 100,000 orders to identify the key factors influencing customer experience. By examining delivery performance, regional logistics, product categories, payment behavior, and review outcomes, the analysis identifies the primary causes of customer dissatisfaction.

The findings indicate that delivery performance relative to the promised date is the strongest determinant of customer satisfaction. Geographic concentration of sellers creates substantial logistics disadvantages for customers in distant regions, while bulky product categories experience additional fulfillment challenges.

The analysis concludes with actionable recommendations focused on dynamic delivery SLAs, seller dispatch performance, regional fulfillment hubs, specialized carrier routing, and transparent payment expectations.

1. Problem Understanding

Olist has experienced significant growth over a two-year period, effectively scaling its order volume and connecting small and medium Brazilian merchants to major e-commerce marketplaces. However, as the platform scaled, customer satisfaction, measured via 1-to-5-star review scores, became inconsistent.

The primary objective of this analysis is to leverage actual operational data—spanning nearly 100,000 orders—to determine what truly drives the customer experience.

By identifying the root causes of customer dissatisfaction and uncovering regional logistics bottlenecks, this report outlines actionable priorities for Olist's leadership to ensure sustainable, high-quality growth.

2. Analytical Approach

To build a comprehensive and accurate view of the business, multiple datasets including orders, payments, products, sellers, customer locations, and reviews were integrated into a unified master dataset.

2.1 Data Cleaning

Non-delivered orders, including canceled and processing orders, were filtered out to accurately measure fulfillment timelines.

Geolocation coordinates were aggregated by zip-code prefix to prevent duplicate records during table joins.

2.2 Metric Engineering

Custom operational metrics were calculated to evaluate fulfillment performance, including:

- **actual_delivery_days**: Total time from purchase to delivery.
- **delay_in_days**: Difference between the estimated delivery date and the actual delivery date.

The general definition of the delivery delay metric is:

$$\text{Delay in Days} = \text{Actual Delivery Date} - \text{Estimated Delivery Date}$$

A negative value indicates that an order arrived early, while a positive value indicates a late delivery.

2.3 Focus Area

The analysis prioritizes evidence-based drivers of satisfaction, specifically:

- Delivery Service Level Agreements (SLAs)
- Regional freight discrepancies (“Logistics Tax”)
- Product category performance
- Payment behavior
- Root causes of poor customer reviews

3. Key Insights & Business Findings

3.1 Finding 1: The Growth vs. Quality Trade-off

Insight:

While Olist successfully scaled its order volume from early 2017 to mid-2018, this rapid expansion strained operational capacity.

During massive demand spikes—most notably the Black Friday surge in November 2017—the average review score sharply declined to **3.96 stars**.

This divergence indicates that carrier infrastructure and seller fulfillment capabilities were likely overwhelmed during peak periods.



Figure 1: Monthly Order Volume vs. Review Score

3.2 Finding 2: The Severe Penalty for Late Deliveries

Insight:

Delivery timing relative to the promised date is the most critical driver of customer satisfaction.

Orders delivered on time or early maintain an exceptional average rating of **4.28 stars**.

However, missing the estimated delivery date creates a significant “Delivery Cliff”:

- On time or early: **4.28 stars**
- Delayed by 1–7 days: **2.70 stars**
- Delayed by more than 7 days: **1.69 stars**

Customers judge delivery performance against their promised expectation, rather than simply considering the absolute number of days taken.

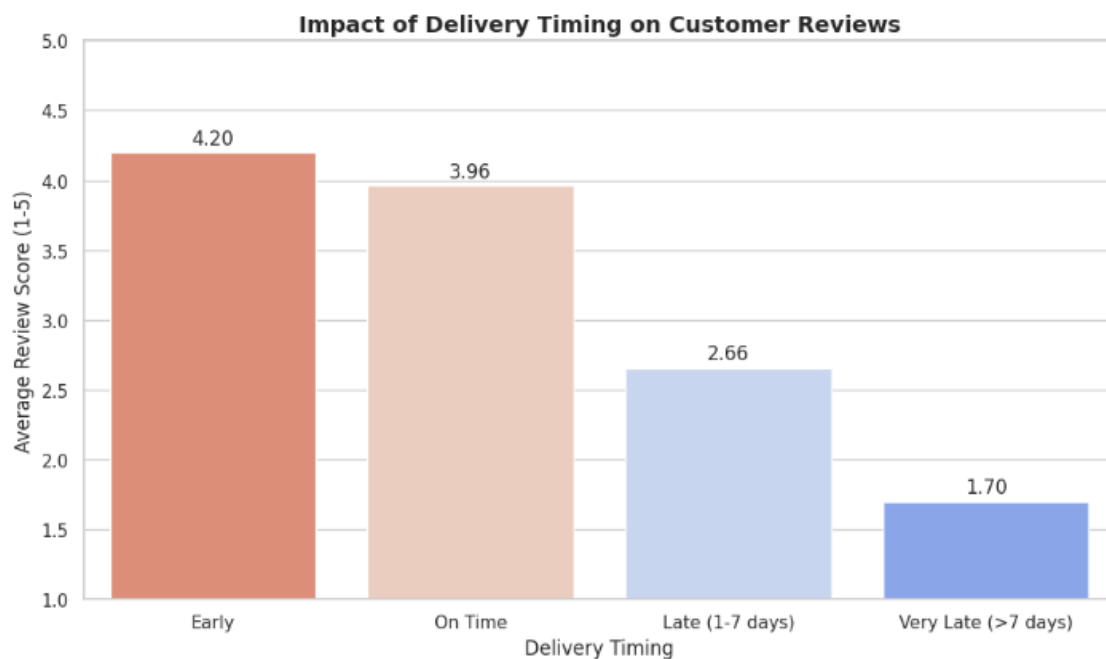


Figure 2: Impact of Delivery Timing on Reviews

3.3 Finding 3: The Geographic Imbalance & “Logistics Tax”

Insight:

The marketplace suffers from severe geographic asymmetry.

A vast majority of sellers are concentrated in the Southeast, particularly around São Paulo. Consequently, buyers in distant regions such as the North and Northeast face a substantial “Logistics Tax.”

Customers in these regions experience:

- Significantly higher freight costs
- Longer transit times
- Average delivery times exceeding **21 days**
- Lower customer satisfaction

The combination of high logistics costs and slow delivery creates a major competitive disadvantage for customers located far from seller clusters.

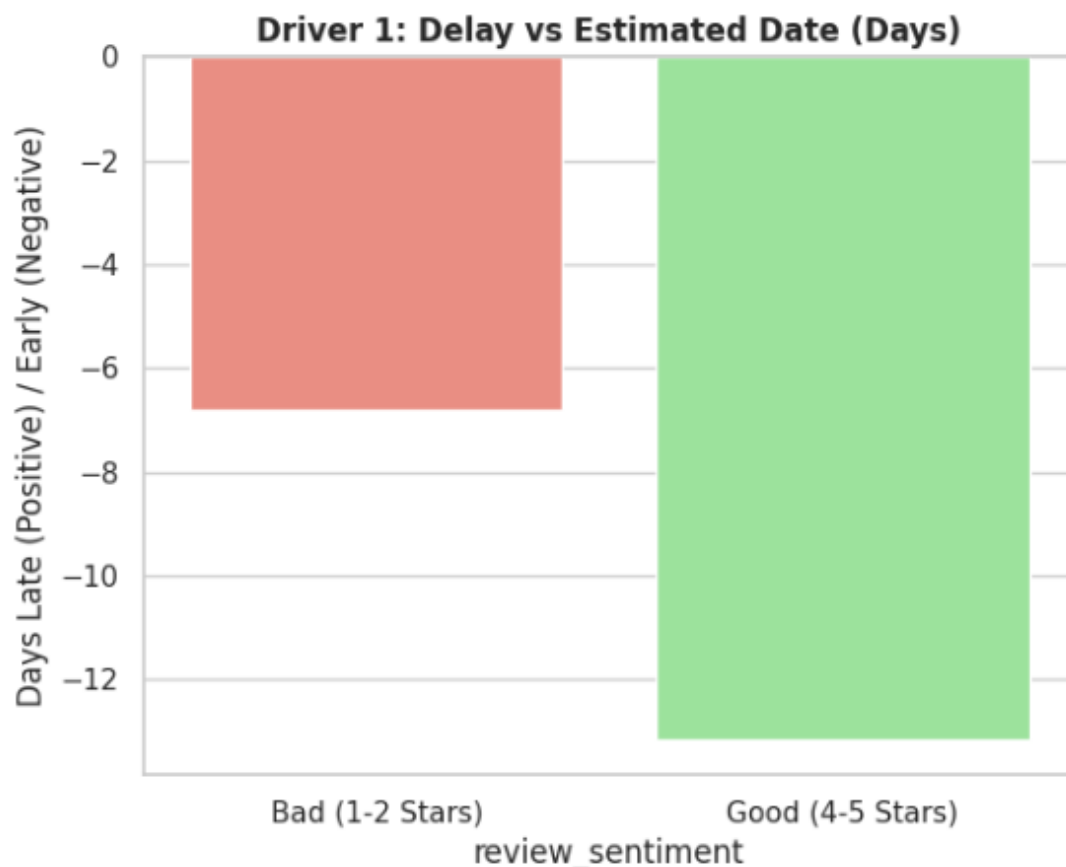


Figure 3: Geographic Inequality: Freight Cost vs. Satisfaction

3.4 Finding 4: The Category Quality Gap

Insight:

Not all product categories are equally suited to the current fulfillment network.

Lightweight and standardized categories such as **Toys** and **Health & Beauty** strongly outperform the platform average, with scores around **4.22 stars**.

Conversely, heavy and bulky categories such as **Bed, Bath & Table** and **Furniture/Decor** consistently underperform.

These categories likely experience higher rates of transit damage or more complex carrier-handling requirements.

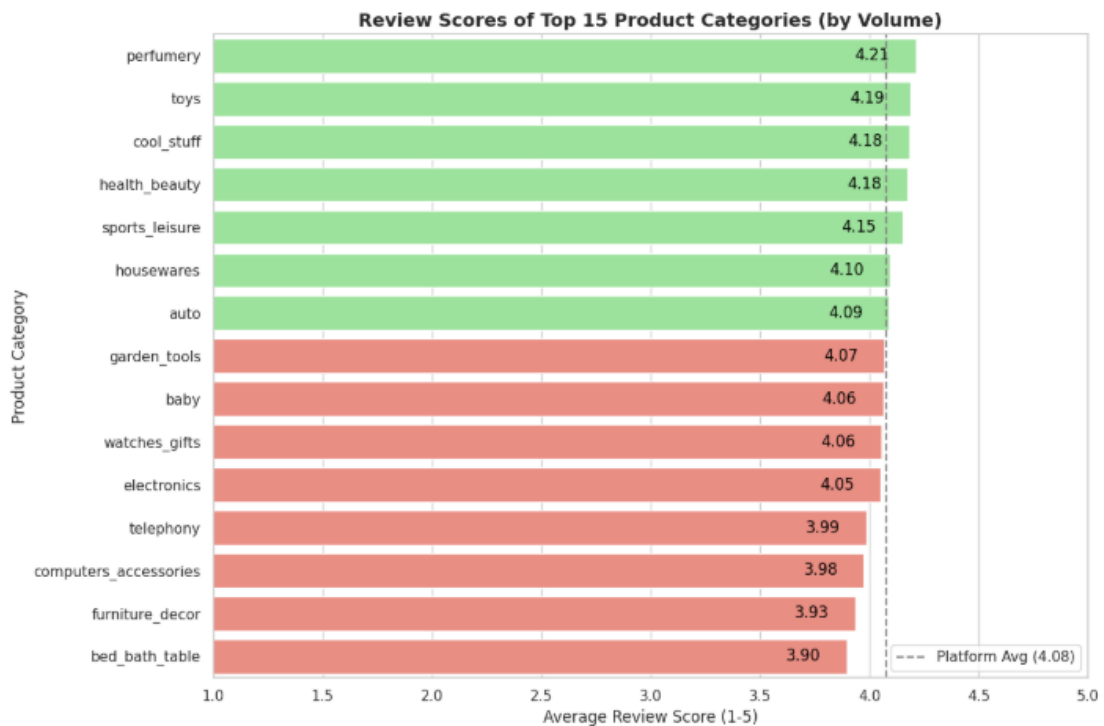


Figure 4: Review Scores of Top 10 Categories

3.5 Finding 5: High-Ticket Friction and Payment Behavior

Insight:

Credit card installment financing is the primary engine for high-value GMV (Gross Merchandise Value), with average order values scaling approximately with the number of installments.

Alternatively, cash-based “Boleto” payments introduce hidden operational friction.

Boletos can take days to clear through the banking system, delaying the seller dispatch trigger and consequently pushing back the final delivery timeline.

This creates a potential mismatch between the customer’s perception of the delivery timeline and the actual operational start of fulfillment.

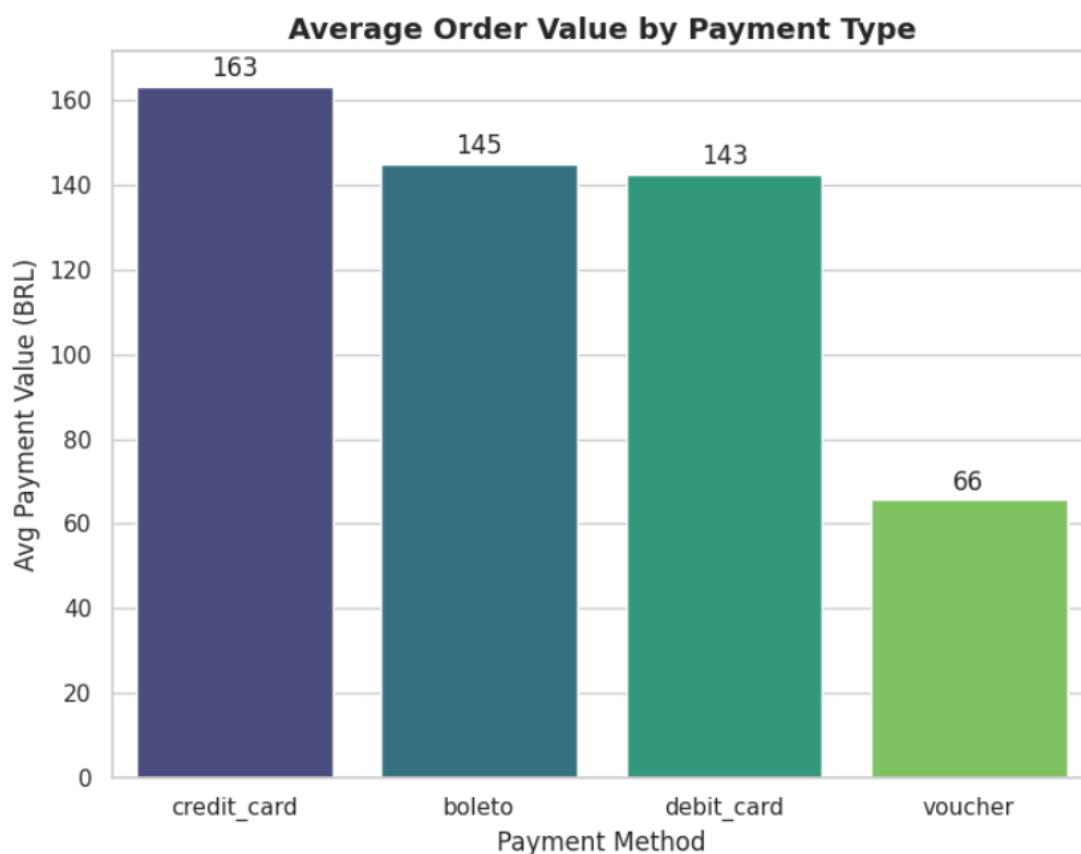


Figure 5: Average Order Value by Payment Type

3.6 Finding 6: Root Cause of Bad Reviews (1–2 Stars)

Insight:

Segmenting 1–2-star reviews from 4–5-star reviews isolates the core drivers of customer dissatisfaction.

The primary root cause is missed delivery expectations.

Orders associated with poor reviews were delivered, on average, **5.1 days past their estimated delivery date**.

In contrast, orders receiving 4–5-star reviews arrived more than **13 days early** on average.

This contrast demonstrates the importance of meeting the customer’s promised delivery date.

Furthermore, when customers pay premium freight costs and still experience SLA failures, the probability of severe dissatisfaction increases substantially.

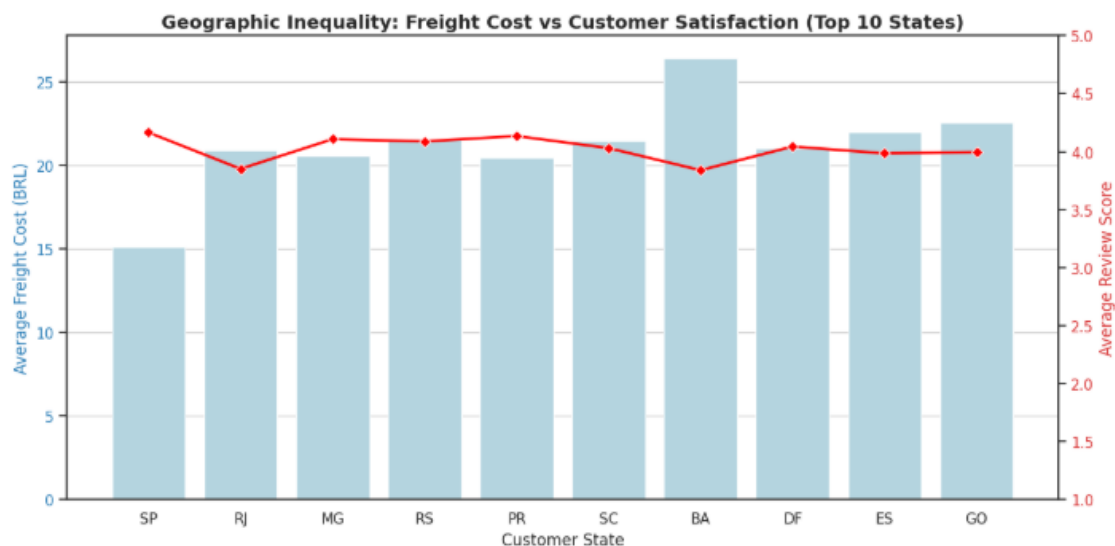


Figure 6: Root Cause: Delay vs. Estimated Delivery

4. Actionable Recommendations

Based on the evidence, Olist leadership should prioritize the following strategic actions.

4.1 1. Dynamic SLA Safety Buffering

Implement a dynamic buffer to customer-facing estimated delivery dates, especially for interstate routes to the North and Northeast.

Under-promising and over-delivering on transit times can improve customer satisfaction without requiring immediate renegotiation of logistics contracts.

The buffer should be data-driven and vary according to:

- Origin-destination pair
- Historical carrier performance
- Product category
- Seasonal demand
- Seller dispatch performance

4.2 2. Enforce Strict Seller Dispatch Policies

A significant portion of delays can originate before the carrier receives the package.

Olist should introduce strict **48-hour seller dispatch policies**, while:

- Penalizing chronic late-dispatch sellers

- Rewarding fast-shipping merchants
- Providing priority listing visibility to reliable sellers
- Monitoring seller-level fulfillment SLAs

This would shift part of the operational responsibility toward sellers and create stronger incentives for reliable fulfillment.

4.3 3. Establish Regional Micro-Hubs

To reduce the “Logistics Tax” for distant buyers, Olist should reduce its reliance on the Southeast as the primary fulfillment origin.

A pilot program could establish regional fulfillment hubs in the Northeast for high-velocity products.

Potential benefits include:

- Lower average freight costs
- Shorter delivery times
- Improved SLA reliability
- Higher regional customer satisfaction
- Better inventory positioning

4.4 4. Specialized Carrier Routing for Bulky Items

Heavy or fragile categories, particularly Furniture and Decor, should not necessarily follow the same logistics network as lightweight parcel products.

Olist should consider partnering with specialized freight networks for bulky categories.

This approach could reduce:

- Transit damage
- Handling-related complaints
- Delivery failures
- Category-specific dissatisfaction

4.5 5. Transparent Payment Expectation Management

Olist should improve checkout communication for customers using **Boleto** payments.

Clear messaging should explain that the delivery estimate begins after the payment-clearing process is completed.

For example, the checkout experience could communicate:

“Your delivery estimate begins after payment confirmation. Boleto payments may require additional processing time before the seller can dispatch your order.”

This would reduce expectation gaps and make the delivery timeline more transparent.

5. Strategic Priorities

The findings suggest that Olist should prioritize customer-experience improvements in the following order:

1. **Delivery SLA Reliability** — The strongest relationship with customer satisfaction.
2. **Seller Dispatch Performance** — Reduce delays before carrier handoff.
3. **Regional Fulfillment** — Address the geographic “Logistics Tax.”
4. **Category-Specific Logistics** — Adapt fulfillment to bulky and fragile products.
5. **Payment Expectation Management** — Reduce friction associated with payment-clearing delays.

6. Conclusion

The analysis demonstrates that Olist’s customer-experience challenge is not primarily a question of growth, but of **operational scalability**.

The most important finding is the sharp relationship between delivery performance and customer satisfaction. Delivering early or on time results in strong customer ratings, whereas even modest delays can trigger a dramatic decline in satisfaction.

Geographic seller concentration further compounds this problem by creating higher freight costs and longer transit times for customers in the North and Northeast. At the same time, bulky product categories face fulfillment challenges that are not experienced to the same extent by lightweight, standardized products.

Therefore, Olist’s next phase of growth should focus not only on increasing order volume, but on building a more resilient fulfillment ecosystem.

The strategic priorities should be to improve SLA accuracy, enforce seller dispatch discipline, decentralize inventory, specialize logistics for difficult product categories, and communicate payment-related delays transparently.

By addressing these operational bottlenecks, Olist can convert its rapid growth into sustainable growth while protecting the customer experience.

A. Chart Reference

The following image files are used in this report. Upload all six images to the same Overleaf project folder as this `main.tex` file.

Table 1: Chart-to-Image Mapping

Chart	Image File
Chart 1	chart1.png
Chart 2	chart2.png
Chart 3	chart3.png
Chart 4	chart4.png
Chart 5	chart5.png
Chart 6	chart6.png

B. Data and Metric Definitions

Table 2: Core Analytical Metrics

Metric	Definition
Actual Delivery Days	Total number of days between the purchase timestamp and actual delivery timestamp.
Delay in Days	Difference between actual delivery date and estimated delivery date.
Review Score	Customer rating on a 1-to-5 star scale.
GMV	Gross Merchandise Value represented by the value of customer orders.
Logistics Tax	Additional freight and transit burden associated with geographic distance from concentrated seller locations.